

Tools for Troubleshooting Theory

1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Here are three real-life scenarios where a multimeter is useful for troubleshooting computers or smartphones:

Computer won't power on – A multimeter measures the power supply's output voltages to verify that the correct DC voltages are reaching the motherboard.

Smartphone not charging – A multimeter checks the charger, USB cable, or charging port voltage to determine whether power is being delivered properly.

Laptop battery suspected to be faulty – A multimeter measures the battery's voltage to see if it is within its expected range or has become discharged or defective.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video

https://youtu.be/PC2zDQ5EFYo?si=w3oMKKtqi5_cp4fq

I watched the “**Motherboard Analyzer Post Code Debugger Overview**” demonstration.

The POST card displays **two-digit hexadecimal codes** that change as the motherboard goes through its startup tests.

In the video, the code changes during startup, allowing the technician to see the different stages, including **DDR5 memory training**.

The card helps troubleshoot **boot/POST problems** by showing where the motherboard stops during testing, pointing the technician toward the faulty hardware.

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

The screenshot shows the HWMonitor application window with the following data:

Sensor	Value	Min	Max
AMD Radeon(TM) Graphics			
Voltages			
GPU	1.420 V	1.319 V	1.430 V
SoC	1.090 V	1.084 V	1.100 V
Temperatures			
GPU	49.0 °C	49.0 °C	55.0 °C
SoC	51.0 °C	50.0 °C	59.0 °C
Powers			
GPU	12.00 W	10.00 W	33.00 W
SoC	3.00 W	3.00 W	4.00 W
Clocks			
Graphics	2200.0 MHz	2200.0 MHz	2200.0 MHz
Memory	2400.0 MHz	2400.0 MHz	2400.0 MHz
Processor	400.0 MHz	400.0 MHz	400.0 MHz
Utilization			
GPU	0.0 %	0.0 %	0.0 %
Video Engine	0.0 %	0.0 %	0.0 %
3D	0.0 %	0.0 %	0.0 %
Copy	0.0 %	0.0 %	0.0 %
Compute	0.0 %	0.0 %	0.0 %
Dedicated Memory	0.3 MB	0.3 MB	0.3 MB
Shared Memory	0.5 MB	0.5 MB	0.5 MB
Memory	0.1 %	0.1 %	0.1 %
Speed			
PCIe Link Speed	16.00 GT/s	16.00 GT/s	16.00 GT/s
AS32-K55			
Voltages			
Current Voltage	16.549 V	16.546 V	16.550 V
Powers			
Charge Rate	0.00 W	0.00 W	0.00 W
Capacities			
Designed Capacity	90006 mWh	90006 mWh	90006 mWh
Full Charge Capacity	63424 mWh	63424 mWh	63424 mWh
Current Capacity	63424 mWh	63424 mWh	63424 mWh
Levels			
Wear Level	29.5 %	29.5 %	29.5 %
Charge Level	100.0 %	100.0 %	100.0 %
VirtualBox Host-Only Ethernet Adapter			
Speed			
Download Speed	0.00 Mbps	0.00 Mbps	0.00 Mbps
Upload Speed	0.00 Mbps	0.00 Mbps	0.00 Mbps
MediaTek Wi-Fi 6 MT7921 Wireless LAN Card			
Speed			
Download Speed	0.01 Mbps	0.00 Mbps	34.10 Mbps
Upload Speed	0.00 Mbps	0.00 Mbps	0.40 Mbps

The screenshot shows the HWMonitor application window with the following data:

Sensor	Value	Min	Max
ASUSTEK COMPUTER INC. FAS07RE			
Temperatures			
LTZ-THRM	66.0 °C	58.0 °C	78.0 °C
Fans			
CPU	4000 RPM	3800 RPM	5000 RPM
Utilization			
Physical Memory Load	62.0 %	62.0 %	65.0 %
Available Physical Me...	5.6 GB	5.3 GB	5.6 GB
Virtual Memory Load	47.9 %	47.9 %	49.6 %
Available Virtual Mem...	16.3 GB	15.7 GB	16.3 GB
AMD Ryzen 7 6800H			
Voltages			
VID (Max)	1.494 V	0.838 V	1.519 V
Temperatures			
Cores (Max)	61.8 °C	51.4 °C	67.9 °C
Package	67.5 °C	57.3 °C	77.3 °C
L3 Cache	59.3 °C	51.7 °C	63.5 °C
GPU	52.8 °C	49.6 °C	55.5 °C
Powers			
Package	15.12 W	7.83 W	41.70 W
Cores	12.16 W	3.77 W	41.66 W
L3 Cache	2.27 W	0.93 W	2.90 W
APU STAPM	18.24 W	18.01 W	18.32 W
APU STAPM Limit	80.00 W	80.00 W	80.00 W
Max Turbo Power (PL2)	80.00 W	80.00 W	80.00 W
Base Power (PL1)	80.00 W	80.00 W	80.00 W
PPT Limit	54.00 W	54.00 W	54.00 W
Levels			
Energy Performance	0.0 %	0.0 %	0.0 %
Clocks			
Cores (Max)	4490.7 MHz	3167.5 MHz	4665.9 MHz
L3 Cache	4170.8 MHz	3895.7 MHz	4399.5 MHz
CPU BCLK	99.8 MHz	99.1 MHz	99.8 MHz
Memory Controller	1197.5 MHz	1188.8 MHz	1198.0 MHz
Memory	2395.2 MHz	2394.2 MHz	2395.9 MHz
Utilization			
Processor	13.4 %	3.2 %	27.3 %
Micron Technology MTC4C10163S15C48BA1 (G591A4F4)			
Voltages			
VDD (Node A)	1.110 V	1.095 V	1.110 V
VDDQ	1.110 V	1.110 V	1.110 V
VPP	1.800 V	1.800 V	1.800 V
VOUT 1.8V	1.815 V	1.800 V	1.815 V
VOUT 1.0V	0.990 V	0.990 V	1.005 V
VIN BULK	5.180 V	5.180 V	5.180 V

4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem Detected
Multimeter	Checks electrical voltage, resistance, and continuity.	Faulty power supply or damaged charging port
POST Card	Checks the motherboard's startup/POST process using diagnostic codes.	RAM or CPU problem preventing the computer from booting
Software Diagnostics	Checks hardware and system performance using diagnostic programs.	Hard-drive errors, overheating, or faulty drivers